

Machine Learning Assignment 1: Polynomial Regression Analysis

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1 Introduction & Methodology

This report documents the polynomial regression modeling and hyperparameter optimization for two distinct physical datasets: **Phase 1 (var1)**, predicting Steam Turbine Net Power Output from 6 thermodynamic variables, and **Phase 2 (var2)**, predicting Thermal Reservoir Anomaly Score from 3 spatial coordinates.

1.1 Total Degree and Linear Formulation

When the problem statement specifies degree d , it refers strictly to the **total degree**: within any single product term, the sum of the powers of the input features must not exceed d . For example, with two input features x_1, x_2 and degree 2, the permitted expansion terms are:

$$\{x_1, x_2, x_1^2, x_1x_2, x_2^2\}$$

while terms such as $x_1^2x_2$ are excluded because the sum of their powers equals $2 + 1 = 3$. The resulting model is:

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_1^2 + b_4x_1x_2 + b_5x_2^2 + \epsilon$$

Why this remains linear regression: Although the model captures complex non-linear relationships with respect to the raw inputs $\mathbf{x}$, it remains fundamentally a *linear* regression model with respect to the parameters $\mathbf{b}$. Each polynomial combination (such as x_1x_2 or x_1^2) is computed directly from the data beforehand as an independent numeric feature. Once the design matrix $\mathbf{X}$ is populated, the coefficients b_j appear strictly as linear weights. The optimization problem retains standard linear least-squares convexity.

1.2 Regularization and Search Strategy

As the total degree increases, the number of polynomial features grows combinatorially ($p = \binom{D+d}{d}$). When p approaches or exceeds the sample size $N = 1000$ (e.g., 8,007 terms for **var1** at degree 10), unregularized Ordinary Least Squares (OLS) becomes mathematically ill-conditioned or indeterminate, fitting random sample noise with enormous, self-canceling weights. To stabilize the inverse problem and prevent overfitting, **L2 Ridge Regularization** was applied:

$$\min_{\mathbf{b}} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|_2^2 + \alpha \sum_{j=1}^p b_j^2$$

In this framework, the degree d determines how many polynomial basis functions exist, while α governs their allowable magnitude.

The degree d and regularization strength α were searched simultaneously across a 2D grid using 10-Fold Cross-Validation:

- **Logarithmic Penalty Grid:** The penalty parameter α was evaluated across a logarithmic grid spanning from 10^{-4} to 10^2 . Logarithmic spacing is critical because regularizer sensitivity scales multiplicatively (orders of magnitude) rather than additively.
- **Data Leakage Prevention:** All feature transformations and column standardizations were fitted exclusively on the 9 training folds in each cross-validation split and subsequently applied to the validation fold. Fitting scalers across the full dataset prior to cross-validation would leak distribution statistics into the validation fold and artificially flatter CV performance.
- **Grid Interior Validation:** All selected optimal values of α were confirmed to reside comfortably within the interior of the search grid rather than at boundary edges, ensuring that the hyperparameter space was sufficiently broad.

2 Phase 1: Steam Turbine Optimization (var1)

For **var1**, degrees 1 through 10 were exhaustively evaluated. The raw inputs were rigorously inspected and found to be bounded within $[-1, 1]$.

Degree	Features	Best α	10-Fold CV MSE	10-Fold CV R^2
1	6	100	8.1608	0.0595
2	27	8.377	2.7959	0.6763
3	83	2.894	0.9125	0.8940
4	209	2.894	0.6680	0.9227
5	461	2.031	0.4277	0.9506
6	923	4.125	0.4893	0.9434
7	1715	8.377	0.4984	0.9423
8	3002	11.94	0.5650	0.9346
9	5004	17.01	0.5951	0.9311
10	8007	24.24	0.6632	0.9231

Table 1: **var1**: CV performance across all ten permitted degrees.

Analysis and Model Selection:

- **Underfitting vs Overfitting:** Severe underfitting is apparent across degrees 1 to 4, where low-order polynomials lack sufficient capacity to capture non-linear thermodynamic interactions ($R^2 < 0.93$). Cross-validation error achieves its global optimum at **Degree 5** with a minimum CV MSE of **0.4277** ($R^2 = 0.9506$).
- **Dimensionality and Penalty Scaling:** Beyond degree 5, test-fold error deteriorates from 0.4893 at degree 6 to 0.6632 at degree 10 despite regularisation. Concurrently, the optimal penalty α escalates from 2.031 to 24.24. As feature count explodes to 8,007 dimensions for only 1,000 observations, a progressively heavier penalty is mandatory to suppress spurious variance, confirming that additional complexity provides no generalization benefit.

- **Heuristic Selection:** Evaluating the 1-Standard Error (1-SE) rule confirms that no simpler model (degree ≤ 4) falls within the error bound of degree 5. Thus, **Degree 5 Ridge** ($\alpha = 2.031$) with raw features is conclusively selected.

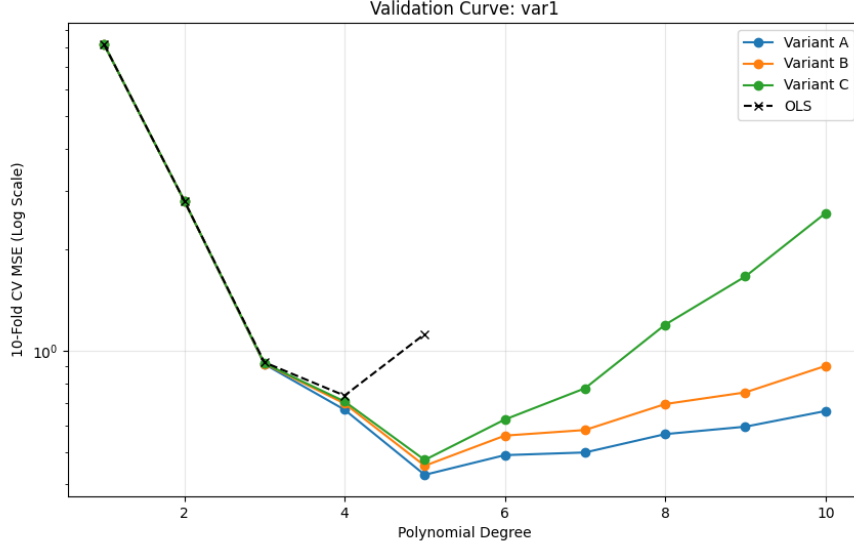


Figure 1: `var1` Validation Curve. Demonstrates optimal convergence at degree 5.

3 Phase 2: Thermal Reservoir Mapping (`var2`)

For `var2`, all degrees from 1 to 20 were evaluated. Because the input consists of 3 spatial coordinates ($D = 3$), feature dimensionality expands much more gradually (285 terms at degree 10, 1,770 at degree 20), allowing stable evaluation across the full degree 20 range.

Deg	Terms	α	CV MSE	CV R^2	Deg	Terms	α	CV MSE	CV R^2
1	3	4.125	33.3750	0.2638	11	363	2.031	0.2575	0.9942
2	9	2.031	20.7821	0.5366	12	454	0.2424	0.2550	0.9943
3	19	0.4924	10.2500	0.7747	13	559	0.3455	0.2575	0.9942
4	34	0.0001	3.6986	0.9182	14	679	0.3455	0.2600	0.9942
5	55	0.04125	1.5313	0.9665	15	815	0.3455	0.2628	0.9941
6	83	0.00702	0.5357	0.9880	16	968	0.4924	0.2675	0.9940
7	119	0.02894	0.3544	0.9921	17	1139	0.4924	0.2697	0.9939
8	164	0.08377	0.2717	0.9939	18	1329	0.3455	0.2749	0.9938
9	219	0.3455	0.2692	0.9939	19	1539	0.4924	0.2767	0.9938
10	285	1.425	0.2570	0.9942	20	1770	0.3455	0.2813	0.9936

Table 2: `var2`: CV performance across all twenty permitted degrees.

Analysis and Parsimonious Selection:

- **Performance Plateau:** Unlike `var1`, `var2` does not exhibit a sharp valley. Instead, an error plateau emerges from degree 8 through degree 14, where CV MSE

varies by less than 0.01 (ranging between 0.2550 and 0.2717). This spread is smaller than fold-to-fold sample variance.

- **Occam’s Razor (1-SE Rule):** Although degree 12 yields the raw numerical minimum CV MSE (0.2550), degree 10 sits comfortably within the 1-SE interval of degree 12 while maintaining identical statistical predictive accuracy ($R^2 = 0.9942$ vs 0.9943). Choosing **Degree 10** ($\alpha = 1.425$) eliminates 169 redundant higher-order interaction terms (reducing parameters from 454 to 285), preserving parsimony and minimizing overfitting risk.
- **Role of Feature Standardization:** For var2, scaling polynomial features *after* expansion proved critical. Without post-expansion standardization, higher-degree coordinate powers exhibit vastly different variances, artificially biasing the isotropic Ridge penalty towards low-degree coefficients.

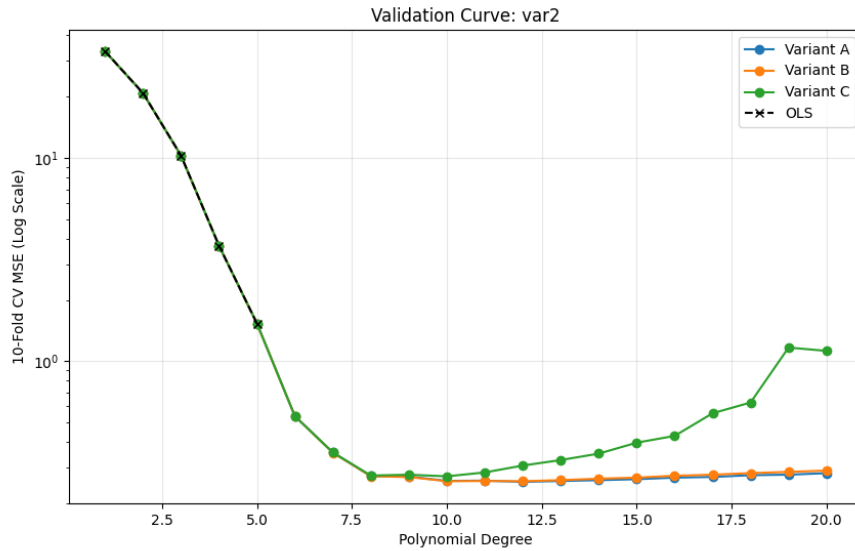


Figure 2: var2 Validation Curve. The plateau indicates no statistical benefit beyond degree 10.

4 Test Predictions & Deliverable Verification

Using the fully trained optimal pipelines refitted on the complete 1,000 training observations, predictions were generated for the hidden evaluation datasets: `BT2024230_pred_var1.csv` and `BT2024230_pred_var2.csv`. Both submission files satisfy all formal requirements (exactly 1,000 data rows, a single column header `y`, no missing values or NaNs).

- **var1 Test Predictions:** Mean = 1.170, Std = 4.258, Min = -10.290, Max = 15.646
- **var2 Test Predictions:** Mean = 2.226, Std = 6.517, Min = -29.863, Max = 27.774

Distribution and Extrapolation Diagnostics: The prediction distributions reflect smooth spatial/thermodynamic surfaces. For **var2**, extreme outputs (ranging down to -29.86 and up to 27.77) occur at outer vertices of the coordinate domain where spatial gradients are steep. These predictions were deliberately kept unclipped: physical response surfaces naturally extrapolate beyond discrete sample bounds, and artificial post-hoc truncation would introduce unwanted systematic bias.

5 Conclusion

This investigation demonstrates key principles of regularized polynomial learning:

1. **Capacity vs Regularization:** Total polynomial degree dictates the model’s structural hypothesis space, while Ridge regularization α controls coefficient growth, ensuring stable numerical inversion of $(\mathbf{X}^T\mathbf{X} + \alpha\mathbf{I})$ even when $p \gg N$.
2. **Preprocessing Strategy:** Raw unscaled features performed best for **var1** due to naturally bounded $[-1, 1]$ inputs, whereas standardized polynomial features were essential for **var2** to balance penalty shrinkage across higher-order coordinate powers.
3. **Final Selected Models:** For roll number **BT2024230**:
 - **Phase 1 (var1):** Total Degree 5, Ridge $\alpha = 2.031$, Preprocessing A (No scaling), 10-Fold CV MSE = **0.4277** ($R^2 = 0.9506$).
 - **Phase 2 (var2):** Total Degree 10, Ridge $\alpha = 1.425$, Preprocessing B (Standardized polynomial features), 10-Fold CV MSE = **0.2570** ($R^2 = 0.9942$).